

RHCSA EX200 — Exambible Exam Dump Q&A

Source: Exambible EX200 (52 questions shown, unique only)

Question 1

Create the user named eric and deny to interactive login.

Solution:

```
useradd eric
passwd eric
vi /etc/passwd
eric:x:505:505::/home/eric:/sbin/nologin
(Change the shell to /sbin/nologin or /bin/false to deny interactive login)
```

Question 2

Install a FTP server, and request to anonymous download from /var/ftp/pub catalog. (it needs you to configure yum direct to the already existing file server.)

Solution:

```
# cd /etc/yum.repos.d
# vim local.repo
[local]
name=local.repo
baseurl=file:///mnt
enabled=1
gpgcheck=0
# yum makecache
# yum install -y vsftpd
# service vsftpd restart
# chkconfig vsftpd on
# chkconfig --list vsftpd
# vim /etc/vsftpd/vsftpd.conf
anonymous_enable=YES
```

Question 3

Set cronjob for user natasha to do /bin/echo hiya at 14:23.

Solution:

```
# crontab -e -u natasha
23 14 * * * /bin/echo hiya
wq!
```

Question 4

One Package named zsh is dump on ftp://server1.example.com under /pub/updates directory and your FTP server is 192.168.0.254. Install the package zsh.

Solution:

```
rpm -ivh ftp://server1.example.com/pub/updates/zsh-*
OR
Login to ftp server: ftp ftp://server1.example.com using anonymous user.
Change the directory: cd pub and cd updates
Download the package: mget zsh-*
Quit from the ftp prompt: bye
Install the package: rpm -ivh zsh-*
Verify: rpm -q zsh
```

Question 5

Configure the verification mode of your host account and the password as LDAP. And it can login successfully through ldapuser40. The password is set as "password".

And the certificate can be downloaded from <http://ip/dir/ldap.crt>. After the user logs on the user has no host directory unless you configure the autofs in the following questions.

Solution:

```
system-config-authentication
LDAP Server: ldap://instructor.example.com (In domain form, not write IP)
OR
# yum groupinstall directory-client (1.krb5-workstation 2.pam-krb5 3.sssd)
# system-config-authentication
1. User Account Database: LDAP
2. LDAP Search Base DN: dc=example,dc=com
3. LDAP Server: ldap://instructor.example.com (In domain form, not write IP)
4. Download CA Certificate
5. Authentication Method: LDAP password
6. Apply
getent passwd ldapuser40
```

Question 6

We are working on /data initially the size is 2GB. The /dev/test0/lvtestvolume is mount on /data. Now you required more space on /data but you already added all disks belong to physical volume. You saw that you have unallocated space around 5 GB on your harddisk. Increase the size of lvtestvolume by 5GB.

Solution:

```
Create a partition having size 5 GB and change the system id to '8e'.
use partprobe command
pvcreate /dev/hda9 (suppose your partition number is hda9)
vgextend test0 /dev/hda9
lvextend -L+5120M /dev/test0/lvtestvolume
```

```
verify using lvdisplay /dev/test0/lvtestvolume
```

Question 7

Configure autofs to make sure after login successfully, it has the home directory autofs, which is shared as /rhome/ldapuser40 at the ip: 172.24.40.10. and it also requires that, other ldap users can use the home directory normally.

Solution:

```
# chkconfig autofs on
# cd /etc/
# vim /etc/auto.master
/rhome /etc/auto.ldap
# cp auto.misc auto.ldap
# vim auto.ldap
ldapuser40 -rw,soft,intr 172.24.40.10:/rhome/ldapuser40
* -rw,soft,intr 172.16.40.10:/rhome/&
# service autofs stop
# service autofs start
# showmount -e 172.24.40.10
# su - ldapuser40
```

Question 8

Download the document from ftp://instructor.example.com/pub/testfile, find all lines containing [abcde] and redirect to /MNT/answer document, then rearrange the order according the original content.

Solution:

```
Download the file to /tmp first
grep [abcde] /tmp/testfile > /mnt/answer
```

Question 9

Create a 2G swap partition which take effect automatically at boot-start, and it should not affect the original swap partition.

Solution:

```
# fdisk /dev/sda
p (check Partition table)
n (create new partition)
Enter
+2G
t
82 (Linux Swap)
w
partx -a /dev/sda
partprobe
mkswap /dev/sda8
```

```
(Copy UUID from output)
swapon -a
vim /etc/fstab
UUID=XXXXX swap swap defaults 0 0
(verbose: swapon -s)
```

Question 10a

Create a backup file named `/root/backup.tar.bz2`, which contains the contents of `/usr/local`, bar must use the bzip2 compression.

Solution:

```
cd /usr/local
tar -jcvf /root/backup.tar.bz2 *
mkdir /test
tar -jxvf /root/backup.tar.bz2 -C /test/
```

Question 10b

Configure a default software repository for your system.

One YUM has already provided to configure your system on `http://server.domain11.example.com/pub/x86_64/Server`, and can be used normally.

Solution:

```
yum-config-manager --add-repo=http://content.example.com/rhel7.0/x86-64/dvd
(generates file: content.example.com_rhel7.0_x86_64_dvd.repo)
Add a line: gpgcheck=0
yum clean all
yum repolist
```

Question 14

Successfully resolve to `server1.example.com` where your DNS server is `172.24.254.254`.

Solution:

```
vi /etc/resolv.conf
nameserver 172.24.254.254
host server1.example.com
```

Question 17

Configure your NFS services. Share the directory by the NFS Shared services.

Solution:

```
/etc/init.d/rpcbind start
/etc/init.d/nfslock start
/etc/init.d/nfs start
chkconfig rpcbind on
chkconfig nfslock on
chkconfig nfs on
```

```
showmount -e localhost
```

Question 20

Find the files owned by harry, and copy it to catalog: /opt/dir

Solution:

```
# cd /opt/  
# mkdir dir  
# find / -user harry -exec cp -rfp {} /opt/dir/ \;
```

Question 23

Configure autofs.

Configure the autofs automatically mount to the home directory of LDAP, as required: server.domain11.example.com use NFS to share the home to your system.

This file system contains a pre configured home directory of user ldapuserX.

Home directory of ldapuserX is: server.domain11.example.com:/home/guests/ldapuser

Home directory of ldapuserX should automatically mount to the ldapuserX of the local /home/guests

Home directory's write permissions must be available for users. ldapuser1's password is password

Solution:

```
yum install -y autofs  
mkdir /home/rehome  
vim /etc/auto.master  
/home/rehome /etc/auto.ldap  
cp /etc/auto.misc /etc/auto.ldap  
vim /etc/auto.ldap  
ldapuserX -fstype=nfs,rw server.domain11.example.com:/home/guests/  
systemctl start autofs  
systemctl enable autofs  
su - ldapuserX (test)  
Note: if command prompt shows -bash-4.2$, there may be a multi-level directory.  
Change to: server.domain11.example.com:/home/guests/ldapuserX
```

Question 26

Configure /var/tmp/fstab Permission.

Copy the file /etc/fstab to /var/tmp/fstab. Configure var/tmp/fstab permissions as the following:

Owner of the file /var/tmp/fstab is Root, belongs to group root

File /var/tmp/fstab cannot be executed by any user

User natasha can read and write /var/tmp/fstab

User harry cannot read and write /var/tmp/fstab

All other users (present and future) can read var/tmp/fstab.

Solution:

```
cp /etc/fstab /var/tmp/  
chmod ugo-x /var/tmp/fstab  
setfacl -m u:natasha:rw- /var/tmp/fstab  
setfacl -m u:harry:--- /var/tmp/fstab  
Verify: getfacl /var/tmp/fstab
```

Question 27

Configure the system synchronous as 172.24.40.10.

Solution:

```
Graphical Interface:  
System --> Administration --> Date & Time  
OR  
# system-config-date
```

Question 32

The user authentication has been provided by ldap domain in 192.168.0.254. According the following requirements to get ldapuser.

-LdapuserX must be able to login your system, X is your hostname number. But the ldapuser's home directory cannot be mounted, until you realize automatically mount by autofs server.

- All ldap user's password is "password".

Solution:

```
system-config-authentication &  
(Configure LDAP settings in the GUI as described in Q5)
```

Question 37

Create a volume group, and set 8M as a extends. Divided a volume group containing 50 extends on volume group lv (lvshare), make it as ext4 file system, and mounted automatically under /mnt/data. And the size of the floating range should set between 380M and 400M.

Solution:

```
# fdisk /dev/vda  
# partprobe  
# pvcreate /dev/vda6  
# vgcreate -s 8M vg1 /dev/vda6  
# lvcreate -n lvshare -l 50 vg1  
# mkfs.ext4 /dev/vg1/lvshare  
# mkdir -p /mnt/data
```

```
# vim /etc/fstab
/dev/vg1/lvshare /mnt/data ext4 defaults 0 0
# mount -a
# df -h
```

Question 38

Please open the `ip_forward`, and take effect permanently.

Solution:

```
vim /etc/sysctl.conf
net.ipv4.ip_forward = 1
sysctl -w (takes effect immediately)
If no sysctl.conf option:
sysctl -a | grep net.ipv4
sysctl -w net.ipv4.ip_forward=1
```

Question 42

Make on data that only the user owner and group owner member can fully access.

Solution:

```
chmod 770 /data
Verify: ls -ld /data
(should show: drwxrwx--- 2 root sysadmin 4096 ...)
```

Question 45

Some users home directory is shared from your system. Using `showmount -e localhost` command, the shared directory is not shown. Make access the shared users home directory.

Solution:

```
Verify the file whether shared or not: cat /etc/exports
Start the nfs service: service nfs start
Start the portmap service: service portmap start
Make nfs start automatically on reboot: chkconfig nfs on
Make portmap start automatically on reboot: chkconfig portmap on
Verify: showmount -e localhost
If firewall is running: iptables -F and stop iptables service
```

Question 46

Add 3 users: `harry`, `natasha`, `tom`.

The requirements: The Additional group of the two users: `harry`, `Natasha` is the admin group. The user: `tom`'s login shell should be non-interactive.

Solution:

```
# useradd -G admin harry
# useradd -G admin natasha
```

```
# useradd -s /sbin/nologin tom
# id harry; id natasha (show additional group)
# cat /etc/passwd (show login shell)
```

Question 48

Search a String

Find out all the columns that contains the string seismic within /usr/share/dict/words, then copy all these columns to /root/lines.tx in original order, there is no blank line, all columns must be the accurate copy of the original columns.

Solution:

```
grep seismic /usr/share/dict/words > /root/lines.txt
```

Question 50

According the following requirements to create a local directory /common/admin.

This directory has admin group.

This directory has read, write and execute permissions for all admin group members.

Other groups and users don't have any permissions.

All the documents or directories created in the /common/admin are automatically inherit the admin group.

Solution:

```
mkdir -p /common/admin
chgrp admin /common/admin
chmod 2770 /common/admin
```

Question 55

Create a swap space, set the size is 600 MB, and make it be mounted automatically after rebooting the system (permanent mount).

Solution:

```
dd if=/dev/zero of=/swapfile bs=1M count=600
mkswap /swapfile
vim /etc/fstab:
/swapfile swap swap defaults 0 0
mount -a
```

Question 59

Upgrading the kernel as 2.6.36.7.1, and configure the system to Start the default kernel, keep the old kernel available.

Solution:

```
# cat /etc/grub.conf
# cd /boot
# lftp it
```



```
# get dr/dom/kernel-xxxx.rpm
# rpm -ivh kernel-xxxx.rpm
# vim /etc/grub.conf
default=0
```

Question 60

Configure a task: plan to run echo hello command at 14:23 every day.

Solution:

```
# which echo
# crontab -e
23 14 * * * /bin/echo hello
# crontab -l (Verify)
```

Question 65

Open kmcrl value of 5 , and can verify in /proc/cmdline

Solution:

```
# vim /boot/grub/grub.conf
Append to the kernel line: kmcrl=5
(Example: ...rhgb quiet kmcrl=5)
Restart, then verify:
# cat /proc/cmdline
```

Question 68

There are two different networks, 192.168.0.0/24 and 192.168.1.0/24. Your System is in 192.168.0.0/24 Network. One RHEL6 Installed System is going to use as a Router. All required configuration is already done on Linux Server. Where 192.168.0.254 and 192.168.1.254 IP Address are assigned on that Server. How will make successfully ping to 192.168.1.0/24 Network's Host?

Solution:

```
vi /etc/sysconfig/network
GATEWAY=192.168.0.254
OR
vi /etc/sysconfig/network-scripts/ifcfg-eth0
DEVICE=eth0
BOOTPROTO=static
ONBOOT=yes
IPADDR=192.168.0.?
NETMASK=255.255.255.0
GATEWAY=192.168.0.254
service network restart
```

Question 71

Create a volume group, and set 16M as a extends. And divided a volume group containing 50 extends on volume group lv, make it as ext4 file system, and mounted automatically under /mnt/data.

Solution:

```
# pvcreate /dev/sda7 /dev/sda8
# vgcreate -s 16M vg1 /dev/sda7 /dev/sda8
# lvcreate -l 50 -n lvm02 vg1
# mkfs.ext4 /dev/vg1/lvm02
# blkid /dev/vg1/lvm02
# mkdir -p /mnt/data
# vim /etc/fstab
UUID=xxxxxxx /mnt/data ext4 defaults 0 0
# mount -a
# mount (Verify)
```

Question 73

Create a 2G swap partition which take effect automatically at boot-start, and it should not affect the original swap partition.

Solution:

```
# fdisk /dev/sda
p (check Partition table)
n (new partition)
Enter
+2G
t
82 (Linux Swap)
w
partx -a /dev/sda
partprobe
mkswap /dev/sda8
(Copy UUID)
swapon -a
vim /etc/fstab
UUID=XXXXX swap swap defaults 0 0
(verify: swapon -s)
```

Question 75

Configure a user account. Create a user iar uid is 3400. Password is redhat

Solution:

```
useradd -u 3400 iar
passwd iar
```

Question 77

Download `ftp://192.168.0.254/pub/boot.iso` to `/root`, and mounted automatically under `/media/cdrom` and which take effect automatically at boot-start.

Solution:

```
# cd /root; wget ftp://192.168.0.254/pub/boot.iso
# mkdir -p /media/cdrom
# vim /etc/fstab
/root/boot.iso /media/cdrom iso9660 defaults,loop 0 0
# mount -a
```

Question 81

Create a volume group, and set the size is 500M, the size of single PE is 16M. Create logical volume named `lv0` in this volume group, set size is 20 PE, make it as `ext3` file system, and mounted automatically under `data`.

Solution:

```
fdisk /dev/vda
pvcreate /dev/vda3
vgcreate -s 16M vg0 /dev/vda3
lvcreate -n lv0 -l 20 vg0
mkfs.ext3 /dev/mapper/vg0-lv0
mkdir /data
vim /etc/fstab:
/dev/mapper/vg0-lv0 /data ext3 defaults 0 0
mount -a
mount | grep data
```

Question 86

The system `ldap.example.com` provides an LDAP authentication service. Your system should bind to this service as follows:

The base DN for the authentication service is `dc=domain11, dc=example, dc=com`

LDAP is used to provide both account information and authentication information.

The connection should be encrypted using the certificate at

`http://host.domain11.example.com/pub/domain11.crt`

When properly configured, `ldapuserX` should be able to log into your system, but will not have a home directory until you have completed the autofs requirement.

Username: `ldapuser11`

Password: `password`

Solution:

```
system-config-authentication
LDAP user DN: dc=domain11,dc=example,dc=com
Server: host.domain11.example.com
Certificate: http://host.domain11.example.com/pub/domain11.crt
```

```
Authentication Method: LDAP password
OK
starting sssd
su - ldapuser11 (verify: displays Bash prompt)
```

Question 90

Create a catalog under /home named admins. Its respective group is requested to be the admin group. The group users could read and write, while other users are not allowed to access it. The files created by users from the same group should also be the admin group.

Solution:

```
# cd /home/
# mkdir admins
# chown .admin admins/
# chmod 770 admins/
# chmod g+s admins/
```

Question 91

Create one partitions having size 100MB and mount it on data.

Solution:

```
1. fdisk /dev/hda (create new partition)
2. Type n (new partition)
3. Press l for logical
4. Press Enter for default start cylinder
5. Type +100M for size
6. Press p to verify, note partition name
7. Press w to write
8. Reboot or run partprobe
9. mkfs -t ext3 /dev/hda?
vi /etc/fstab:
/dev/hda? /data ext3 defaults 1 2
mount /dev/hda? /data (verify current session)
```

Question 94

Configure the permissions of /var/tmp/fstab

Copy the file /etc/fstab to /var/tmp/fstab. Configure the permissions of /var/tmp/fstab so that:

the file /var/tmp/fstab is owned by the root user.

the file /var/tmp/fstab belongs to the group root.

the file /var/tmp/fstab should not be executable by anyone.

the user natasha is able to read and write /var/tmp/fstab.

the user harry can neither write nor read /var/tmp/fstab.

all other users (current or future) have the ability to read /var/tmp/fstab.

Solution:

```
cp -a /etc/fstab /var/tmp
cd /var/tmp
ls -l
getfacl /var/tmp/fstab
chmod ugo-x /var/tmp/fstab
setfacl -m u:natasha:rw /var/tmp/fstab
setfacl -m u:harry:0 /var/tmp/fstab
Verify: ls -la /var/tmp/fstab
```

Question 96

Search files. Find out files owned by jack, and copy them to directory /root/findresults

Solution:

```
mkdir /root/findfiles
find / -user jack -exec cp -a {} /root/findfiles/ \;
ls /root/findresults
```

Question 100

One Logical Volume named /dev/test0/testvolume1 is created. The initial Size of that disk is 100MB now you required more 200MB. Increase the size of Logical Volume, size should be increase on online.

Solution:

```
lvextend -L+200M /dev/test0/testvolume1
lvdisplay /dev/test0/testvolume1
ext2online -d /dev/test0/testvolume1
```

Question 103

Make a swap partition having 100MB. Make Automatically Usable at System Boot Time.

Solution:

```
fdisk /dev/hda (create new partition)
Type n (new partition)
Press l for logical
Press Enter for default start cylinder
Type +100M
Press p to verify
Type t to change system ID
Type 82 (Linux Swap)
Press w to write
Reboot or run partprobe
mkswap /dev/hda?
```

```
swapon /dev/hda?  
free -m (verify)  
vi /etc/fstab:  
/dev/hda? swap swap defaults 0 0  
Reboot and verify swap is automatically enabled
```

Question 104

One Logical Volume is created named as myvol under vo volume group and is mounted. The Initial Size of that Logical Volume is 400MB. Make successfully that the size of Logical Volume 200MB without losing any data. The size of logical volume 200MB to 210MB will be acceptable.

Solution:

```
First check the size: lvdisplay /dev/vo/myvol  
Check filesystem consistency: fsck -f /dev/vo/myvol  
Reduce the filesystem: resize2fs /dev/vo/myvol 200M  
Reduce the logical volume: lvreduce /dev/vo/myvol -L 200M  
Verify size: lvdisplay /dev/vo/myvol  
Verify online: df -h
```

Question 105

There is a local logical volumes in your system, named with shrink and belong to VGSRV volume group, mount to the /shrink directory. The definition of size is 320 MB.

Requirement:

Reduce the logical volume to 220 MB without any loss of data. The size is allowed between 200-260 MB after reducing.

Solution:

```
cd; umount /shrink  
e2fsck -f /dev/mapper/vgsrv-shrink  
resize2fs /dev/mapper/vgsrv-shrink 220M  
lvreduce -L 220M /dev/mapper/vgsrv-shrink  
mount -a
```

Question 108

There is a server having 172.24.254.254 and 172.25.254.254. Your System lies on 172.24.0.0/16. Make successfully ping to 172.25.254.254 by Assigning following IP: 172.24.0.x where x is your station number.

Solution:

```
Use netconfig command  
Enter the IP Address (example: 172.24.0.1)  
Enter Subnet Mask  
Enter Default Gateway and primary name server  
Press ok
```

```
ifdown eth0
ifup eth0
verify using ifconfig
```

Question 110

SELinux must be running in the Enforcing mode.

Solution:

```
getenforce (check current mode)
If not enforcing:
vim /etc/selinux/config
SELINUX=enforcing
setenforce 1 (temporary, takes effect immediately)
sestatus (verify)
```

Question 111

Binding to an external validation server.

System server.domain11.example.com provides a LDAP validation service, your system should bind to this service as required:

Base DN of validation service is dc=example,dc=com

LDAP is used for providing account information and validation information

Connecting and using the certification of

<http://server.domain11.example.com/pub/EXAMPLE-CA-CERT> to encrypt

After the correct configuration, ldapuser1 can log into your system, it does not have HOME directory until you finish autofs questions, ldapuser1 password is password.

Solution:

```
yum -y install sssd authconfig-gtk krb5-workstation
authconfig-gtk (open graphical interface)
Modify user account database to LDAP
Fill in DN and LDAP SERVER as required
Check "Use TLS to encrypt connections"
Download CA cert: http://server.domain11.example.com/pub/EXAMPLE-CA-CERT
Authentication method: LDAP password
Test: id ldapuser1
Note: user password does not need to be set
```